

Chapter Three

Information Security Techniques

Part – I

Classical Encryption Principles & Algorithms

Information Security (SE7431)

INTRODUCTION

- ✎ Institutions of all sizes collect and store huge volumes of confidential information and most of this information is collected, processed and stored on computers and transmitted across networks to other computers.
- ✎ The rapid growth and widespread use of electronic data processing and electronic business conducted through the Internet, along with numerous occurrences of international terrorism, fueled the need for better methods of protecting the computers and the information.
- ✎ Information security means protecting information and information systems from unauthorized access, use, disruption, or destruction.
- ✎ So that, the OSI **security architecture** provides a systematic frame work for defining security attacks

TERMINOLOGIES

Cryptography:

- The art of protecting transmitted information from unauthorized interception or tampering
- The art and science of encompassing the principle and methods of transforming an intelligible (plain text) message into one that is unintelligible(cipher text), and then retransforming that message back to its original form.
- The sender enciphers a message into unintelligible form, and the receiver deciphers it into intelligible form
- The enciphering and deciphering of messages into secret codes by means of various transformations of the plaintext
- Cryptography is closely related to communication theory, namely coding theory
- Coding Theory involves translating information of any kind (text, scientific data, pictures, sound, and so on) into a standard form for transmission, and protecting this information against distortion by random noise.

TERMINOLOGIES

Plain text:

- ☞ The original intelligible message or data that is fed into the algorithm as input
- ☞ The plaintext is not quite the same as the message being sent
- ☞ The message probably has to be translated into some standard form to be encrypted; for example, this might be leaving out the punctuation, turning it into ASCII code or a sequence of numbers, etc.
- ☞ But there is nothing secret about this stage; knowing the plaintext is equivalent to knowing the message

Cipher text:-

- ☞ The scramble message produced as an output of the encryption algorithm

Key(Secret Key):

- ☞ Is some critical information used by the encryption algorithm as input to encrypt the plaintext
- ☞ It is independent of the plaintext and encryption algorithm
- ☞ It is only known to the sender and receiver

TERMINOLOGIES

✚ Enciphering or Encryption:

- ✚ The process of converting the plaintext to cipher text
- ✚ Considered as a component of the cryptographic framework
- ✚ It's role is to offer the confidentiality axis of the CIA triad
- ✚ Needs an input (Clear message & key) to deliver the cipher form (output) , this cipher form to be decrypted (converted to the clear form) we shall need a key and the same algorithm

✚ Deciphering/Decryption:- is the process of restoring back the plaintext from the cipher text.

✚ Cryptanalysis(code breaking):-The process of deriving the plaintext from the cipher text (breaking a code) without being in possession of the key or the system

✚ Cryptology:-The scientific study of cryptography and cryptanalysis

APPROACHES TO INFORMATION SECURITY

Link Encryption:

- ☞ Each vulnerable communication link is equipped on both ends with an encryption device.
- ☞ *User information, header, trailers, source / destination addresses and routing data will be encrypted*
- ☞ The only data/information that will not be encrypted is data link control message information
- ☞ The main disadvantage is that it is effective only if all potential weak links from source to destination are secured.
- ☞ It is designed to protect packet sniffers and eavesdroppers

End-to-End Encryption:

- ☞ Data is encrypted only at the source node and decrypted at the destination node.
- ☞ Nobody in between the sender and the receiver
- ☞ The cryptographic keys used to encrypt and decrypt the messages are stored exclusively on the endpoints

CHARACTERISTICS OF CRYPTOGRAPHIC SYSTEMS

✧ All encryption algorithms are based on two general principles:

- ✧ Substitution: Element in the plain text(bit, letter or groups) is mapped into another element
- ✧ Transposition: Elements in the plain text are rearranged

✧ The number of keys used:-

- ✧ Symmetric: If both sender and receiver use similar or single key for both encryption and decryption, then its called conventional encryption
- ✧ Asymmetric: If the sender and receiver use different keys or two keys, then its called Public-key encryption.

✧ Processing plain text

- ✧ Block cipher: Processes the input one block of elements(typically 64 or 128 bits) at a time, producing an output block for each input block
- ✧ Stream cipher: Processes the input elements (one bit/byte) continuously, producing output one element at a time, as it goes along

SYMMETRIC (CONVENTIONAL) ENCRYPTION

- ✚ The algorithm should be strong enough so that an opponent who knows the algorithm and has access to one or more cipher text would be unable to decipher the cipher text or figure out the key.
- ✚ Sender & receiver must have to obtain the secret key in secure fashion and keep it secure
- ✚ If someone can found the key and knows the algorithm, all communication using the key is readable
- ✚ If the key is generated at the message source, then it must be provided to the destination by means of some secure channel
- ✚ Alternatively, a third party could generate the key and securely deliver it to both source and destination

ATTACKING SYMMETRIC ENCRYPTION

✧ Cryptanalysis:

- ✧ To recover the plaintext of a cipher text or, more typically, to recover the secret key
- ✧ A method for circumventing the security of a cryptographic system by finding a weakness in a code, cipher, cryptographic protocol or key management scheme
- ✧ Rely on nature of the algorithm and some knowledge of plaintext characteristics and even some sample plain text - cipher text pairs
- ✧ Exploits characteristics of algorithm to deduce specific plaintext or key
- ✧ Kerkhoff's principle: The adversary knows all details about a cryptosystem except the secret key

✧ Brute-force Attack:

- ✧ Try all possible keys on cipher text until get an intelligible translation into plaintext
- ✧ On average, half of all possible keys must be tried to achieve success

✧ Cryptanalytic Attack: Different forms of cryptanalysis

CRYPTANALYTIC ATTACK

✧ Classified by how much information needed by the attacker:

✧ Cipher text-only attack, Known-plaintext attack, Chosen-plaintext attack, Chosen-cipher text attack and Chosen-text attack

✧ Cipher text-only attack

✧ Attacker knows cipher text of several messages encrypted with same key

✧ Possible to recover plaintext (also possible to deduce key) by looking at frequency of cipher text letters

✧ That is given a cipher text c and the question is, what is the plaintext m ?

✧ **An encryption scheme is completely insecure if it cannot resist cipher text-only attacks**

✧ The easiest to defend against because the opponent has the least amount information to work with

CRYPTANALYTIC ATTACK

Known-plaintext attack

- ☞ The attacker knows the Algorithm, cipher text and plaintext-cipher text pair
- ☞ Attackers observe pairs of plaintext-cipher text encrypted with the same secret key.
- ☞ Possible to deduce key and/or devise algorithm to decrypt cipher text.
- ☞ That is given $(m_1, c_1), (m_2, c_2), \dots, (m_k, c_k)$ and a new cipher text c
- ☞ Q: what is the plaintext m and what is the secret key in use?

Chosen-plaintext attack

- ☞ Attacker can choose the plaintext and look at the paired cipher text
- ☞ Attacker has more control than known-plaintext attack and may be able to gain more information about key
- ☞ That is given: $(m_1, c_1), (m_2, c_2), \dots, (m_k, c_k)$, where $m_1, m_2, \dots, m_k$ are chosen by the adversary; and a new cipher text c
- ☞ Q: what is the plaintext c , or what is the secret key?
- ☞ The attacker knows the Algorithm, cipher text and chosen plaintext and its cipher text

CRYPTANALYTIC ATTACK

✚ Chosen-cipher text attack

- ✚ Attacker chooses a series of plaintexts, basing the next plaintext on the result of previous encryption
- ✚ The Encryption algorithm, cipher text and purported cipher text chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key are known to cryptanalyst.
- ✚ That is given: $(m_1, c_1), (m_2, c_2), \dots, (m_k, c_k)$, where $c_1, c_2, \dots, c_k$ are chosen by the adversary; and a new cipher text c .
- ✚ Q: what is the plaintext of c , or what is the secret key?
- ✚ The attacker knows the Algorithm, cipher text & chosen plaintext and its cipher text

CRYPTANALYTIC ATTACK

✧ Chosen-text attack

- ☞ The attacker has the capability to choose arbitrary plaintexts to be encrypted and obtain the corresponding cipher texts
- ☞ The attacker knows the Algorithm, cipher text, chosen plaintext and chosen cipher text
- ☞ The goal of the attack is to gain some further information which reduces the security of the encryption scheme.
- ☞ In the worst case, a chosen-plaintext attack could reveal the scheme's secret key.
- ☞ Two forms of Chosen-text attack
 - ☞ Batch chosen-plaintext attack - The attacker chooses all plaintexts before any of them are encrypted
 - ☞ Adaptive chosen-plaintext attack: The attacker has the ability to make his choice of the inputs to the encryption algorithm based on the previous chosen plaintext queries and their corresponding ciphertexts

TRANSPOSITION CIPHER

- ✚ Rearrange letters in plaintext to produce cipher text
- ✚ Rail-Fence Cipher, Route Cipher, Columnar transposition, Double transposition, Disrupted Transposition and Myszkowski Transposition

✚ Rail-Fence Transposition:

- ✚ The plaintext is written downwards and diagonally on successive "rails" of an imaginary fence
- ✚ Move up when we reach the bottom rail and move down when we reach the top rail
- ✚ The message is then read off in rows.
- ✚ Example: If we have 3 "rails" and a message of '**KILL THE QUEEN**', the cipher writes out
- ✚ Encryption/Transposition Result: **KTU ILHQEN LEE**

K				T				U			
	I		L		H		Q		E		N
		L				E				E	

TRANSPOSITION CIPHER

✧ Route Cipher:

- ✧ In a route cipher, the plaintext is first written out in a grid of given dimensions, then read off in a pattern given in the key.
- ✧ Route ciphers have many more keys than a rail fence.
- ✧ In fact, for messages of reasonable length, the number of possible keys is potentially too great to be enumerated even by modern machinery.
- ✧ Example, if we have 3 "rails" and a message, “**KILL THE QUEEN**”, the cipher writes out:
- ✧ The key might specify “**spiral inwards, clockwise, starting from the top right**”
- ✧ Encryption/Transposition Result: **EENUHLIK LEQT**

K	L	E	E
I	T	Q	E
L	H	U	N

TRANSPOSITION CIPHER

✧ Columnar transposition

- ✧ The message is written out in rows of a fixed length, and then read out again column by column, and the columns are chosen in some scrambled order.
- ✧ Both the width and the permutation of the columns are usually defined by a keyword.
- ✧ Example, the word ZEBRAS is of length 6 (so the columns are of length 6), and the permutation is defined by the alphabetical order of the letters in the keyword.
- ✧ In this case, the order would be "6 3 2 4 1 5".
- ✧ In a regular columnar transposition cipher, any spare spaces are filled with nulls; in an irregular columnar transposition cipher, the spaces are left blank
- ✧ Example: **WE ARE DISCOVERED FLEE AT ONCE**
- ✧ Encryption/Transposition Result:

EVLNF**ACDT**S**ESA**A**ROFOD**D**EEC**Q**WIREE**

6	3	2	4	1	5
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	A	S	D	F	Q

TRANSPOSITION CIPHER

✧ Double transposition:

- ✧ A single columnar transposition could be attacked by guessing possible column lengths, writing the message out in its columns and then looking for possible anagrams.
- ✧ Thus to make it stronger, a double transposition was often used, this is simply a columnar transposition applied twice.
- ✧ The same key can be used for both transpositions, or two different keys can be used.
- ✧ Example: Take the result of the irregular columnar transposition in the previous section, and perform a second encryption with a keyword, STRIPE, which gives the permutation "564231"

✧ Message: **EVLNFACDTSESEAROFODDEECQWIREE**

✧ Encryption/Transposition Result:

ASFCE NSRER FEOEE LTADI ECEOQVDADW

5	6	4	2	3	1
E	V	L	N	F	A
C	D	T	S	E	S
E	A	A	R	O	F
O	D	D	E	E	C
Q	W	I	R	E	E

EXAMPLE

✧ We are software Engineers at WKU.

✧ 1. Soft

✧ 2. Ware

✧ 1. Construct columns

✧ 2. Orders to read columns

4	1	3	2
A	O	A	N
E	A	U	E
S	W	E	N
S	K	W	E
T	E	I	R
W	R	F	R
G	E	T	

✧ $K_1 \Rightarrow C_1 = \text{AOANEAU ESWENSK WETEIRW RFRGET}$

✧ $K_2 \Rightarrow C_2 = \text{OAWKERE NENERR AUEWIFT AESSTWG}$

TRANSPOSITION CIPHER

✧ Myszkowski Transposition

- ✧ A variant form of columnar transposition THAT requires a keyword with recurrent letters.
- ✧ In usual practice, subsequent occurrences of a keyword letter are treated as if the next letter in alphabetical order, e.g., the keyword TOMATO yields a numeric key string of "532164."
- ✧ In Myszkowski transposition, recurrent keyword letters are numbered identically, TOMATO yielding a key string of "432143".
- ✧ Example: Message – WE ARE DISCOVERED FLEE AT ONCE.
- ✧ Encryption/Transposition Result:

ROFOD ACDTS *EDSEEEACAQWEIVRLENEF*

4	3	2	1	4	3
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	A	S	D	F	Q

TRANSPOSITION CIPHER

Reading Assignments and Exercises

Reading Assignment:

-  Read about Disruptive columnar Transposition Cipher: About History, Application, Algorithm and related issues.

Exercises:

-  Implement all transposition encryption and decryption algorithms discussed in this course material with any programming language, a programming language you prefer to implement.
-  Note: All algorithms should be implemented with one programming language. You are supposed to include disruptive columnar transposition encryption algorithm in your project.
-  Submission Date: As soon as chapter 3 is discussed and completed